

Bambu Labs X1 - Carbon



Specs.

- 10x10x10in Build Plate
- Ability to Print 4 Materials
- Dual auto bed leveling
- AI inspected first layer
- Max Hot End Temperature - 300C

3D Printed Functional Parts Being Used

Advantages to owning a 3D Printer

- Rapid Prototyping
- Lowers Cost
- Lowers Lead Times
- More Control On Parts



VSI Cost Savings Prototypes



Hand Controller Test Fitting

Cost Savings

VSI Quick Connect

Printing Cost: \$3.50
 Proto Labs Cost: \$180.85
 Printing Time: 4 HR 11 MIN



Hand Controller Housings

Printing Cost: \$1.30
 Proto Labs Cost: \$96.82
 Printing Time: 2 HR 15 MIN



VSI Prototype Piping

Printing Cost: \$19.08
 Proto Labs Cost: \$682.27
 Printing Time: 20 HR



Materials We Can Print



PLA PROS

- Easy to Print
- Fast to Print

CONS

- Very Brittle



ABS - GF PROS

- Durable
- Water Resistance

CONS

- Warping
- Hygroscopic



PAHT-CF PROS

- High Tensile Strength
- Low Moisture Absorption

CONS

- Warping



TPU 95A HF PROS

- Fast to Print
- Flexible
- Layer Adhesion

CONS

- Stringing



PC PROS

- Water Resistance
- Chemical Resistance

CONS

- Hygroscopic



PA6-CF PROS

- Stiffness
- Toughness

CONS

- Hygroscopic
- Warping

Full list of materials and properties can be found on [Bambu Labs](https://www.bambu-labs.com/)